

Hexatic Molecular Orientational Order in a 2D Supercooled Liquid

Leopold Li and Richard M. Stratt



Introduction

Hexatic *molecular-orientational* order is observed to emerge in a deeply supercooled regime of the two-dimensional Lewis-Wahnström (2dLW) model of Ortho-terphenyl, a known glass-former. This ordering is mesoscopic, persists towards the glass transition, and exhibits a character dependent on the cooling rate. While similar orientational ordering has been demonstrated in liquids of rigid polygons, why a preference for hexatic molecular orientational ordering exists in 2dLW is not obvious. To better understand the role of the observed order in the dynamics of 2dLW, we use molecular dynamics simulations to study the behavior of 2dLW at a range of temperatures and explore Monte-Carlo techniques towards accelerating relaxation at low temperatures. We compute measures of liquid structure and closely examine the phase transition towards resolving mechanisms responsible for the onset of observed order. Finally, analytical calculations using Scaled-Particle Theory using a hard-sphere variant of 2dLW suggest that a preference for hexatic order is a result of molecular geometry.

The 2D Lewis-Wahnström Model

The Lewis-Wahnström model replaces each phenyl ring in Ortho-terphenyl with a Lennard-Jones potential centered at a single interaction site. We study a two-dimensional variant (2dLW) with a 75° apical angle.

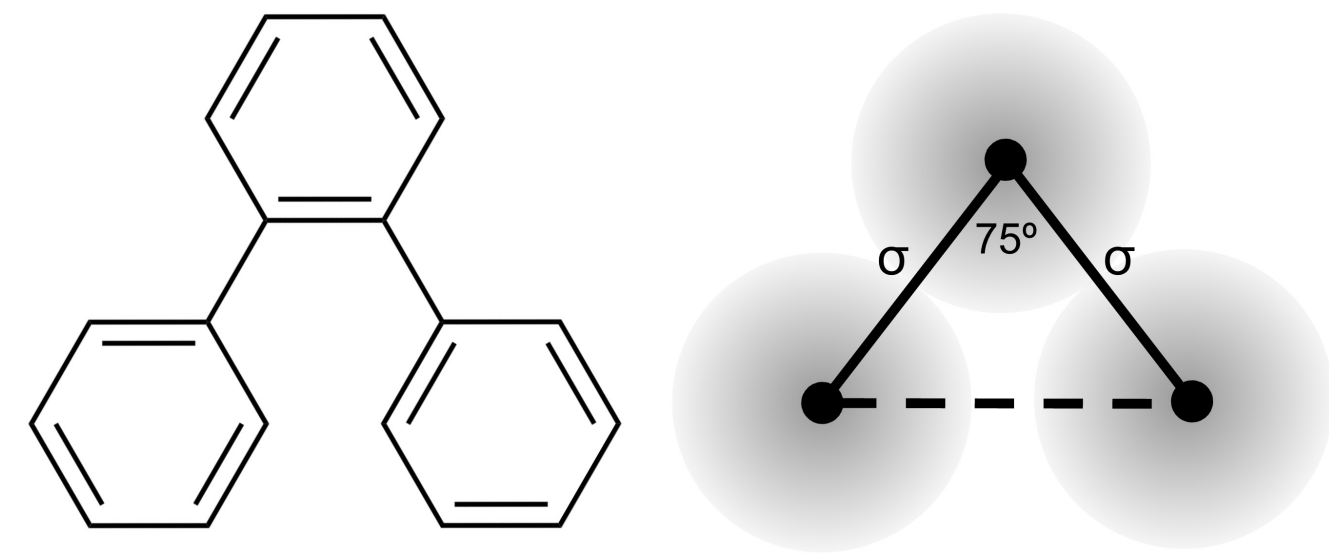
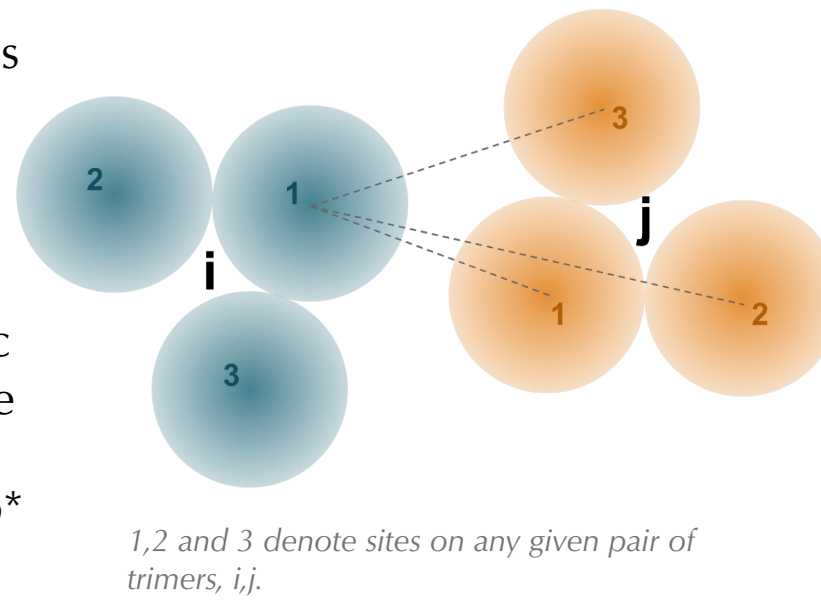


Figure 1: Skeleton model of Ortho-terphenyl (left) and the 2dLW model (right). Each circle represents a Lennard-Jones site. Lines denote rigid bonds, and the dashed line is a fictional bond used to maintain the apical angle. Here, the reduced unit σ represents a separation of 4.83 Å.

Molecular Dynamics

We simulate the dynamics of a 2dLW liquid using molecular dynamics simulations in the microcanonical ensemble. We approximate the Hamiltonian of our system as the sum of total kinetic energy and pairwise potentials. Trajectories are obtained using velocity Verlet integration with a timestep of 0.001τ (in reduced Lennard-Jones time units). We enforce the geometry of 2dLW molecules using holonomic constraints via Lagrange multipliers calculated using an altered routine from the popular RATTLE algorithm [1]. The simulation domain was a square box of dimensions determined by a fixed reduced density of $\rho^* = 0.25$, and periodic boundary conditions. More recently, we have developed molecular dynamics simulations in the canonical and isothermal-isobaric ensembles using a measure-preserving algorithm based on the Martyna-Tobias-Klein equations of motion [7] using Nosé-Hoover chain thermostats in order to simulate a system without cavitation.



Molecular Orientational Order

Molecular-orientational order, not to be confused with Bond-orientational order, describes the alignment of molecular symmetry axes with some common director. Orientational order can be described by the order parameter tensor (a matrix in 2D), $\underline{Q}^{(m)}$, and the m -fold orientational order parameter, $|s_m|$. In an isotropic liquid, $|s_m| = 0$, while in a perfectly ordered liquid, $|s_m| = 1$.

$$\underline{Q}^{(m)} = \frac{1}{N} \sum_i \begin{bmatrix} \cos(m\theta_i) & \sin(m\theta_i) \\ \sin(m\theta_i) & -\cos(m\theta_i) \end{bmatrix}$$

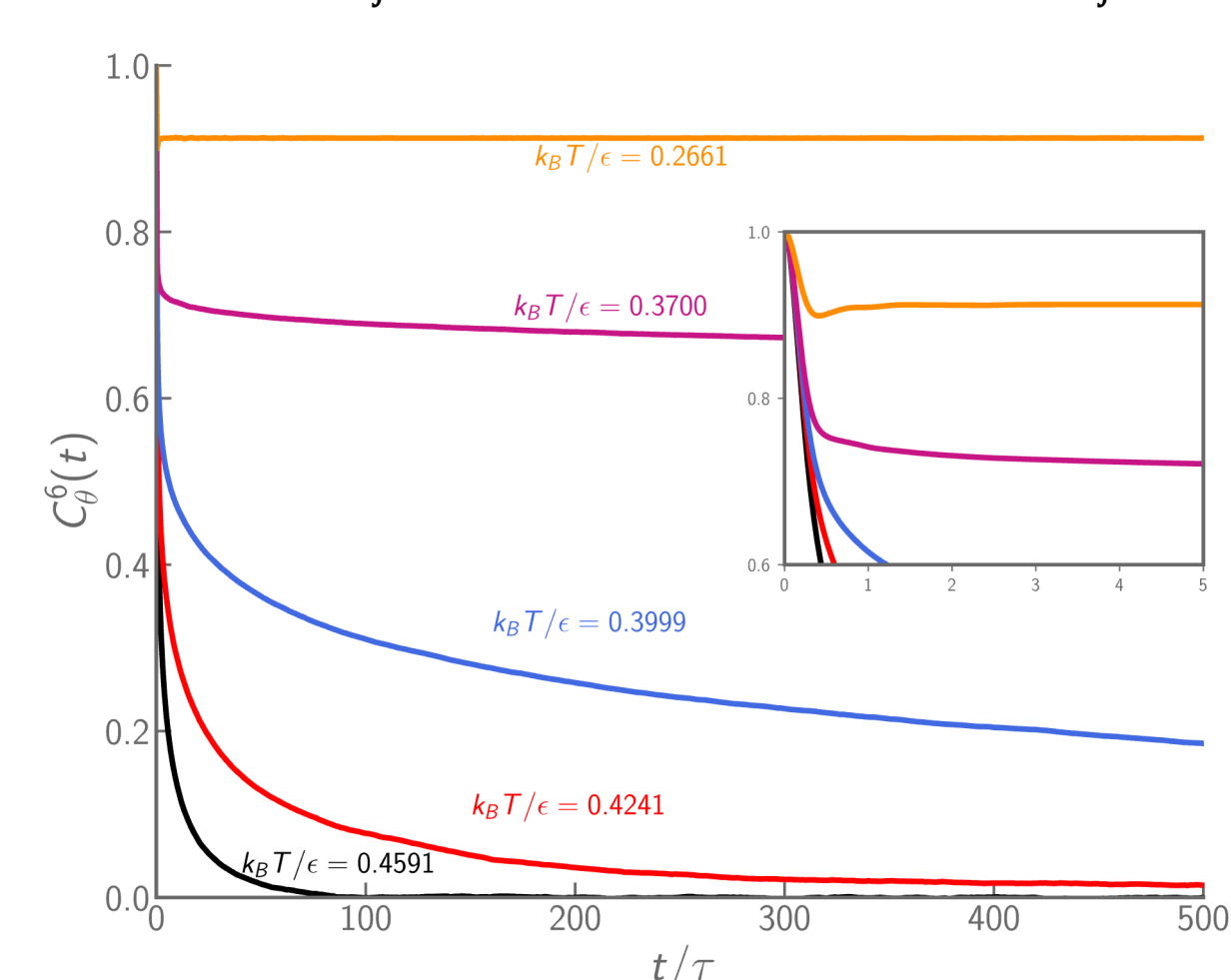
$$\langle |s_m| \rangle = \left\langle \left| \frac{1}{N} \sum_j e^{im\theta_j} \right| \right\rangle_t$$

Figure 2: Schematic of nematic (left) and isotropic (right) phases of an ellipsoidal liquid crystal. Here, the nematic phase exhibits twofold order with $|s_2| = 1$, while the isotropic phase has no such order.

Slow Dynamics

Supercooled liquid dynamics are very slow. Time correlation functions show the timescale that liquids retain transient properties. Here, sixfold orientation becomes increasingly long-lived with cooling.

$$C_\theta^{(m)}(t) = \left\langle \frac{1}{N} \sum_j \mathbf{T}_m(\hat{\Omega}_j(t_0 + t) \cdot \hat{\Omega}_j(t_0)) \right\rangle_t = \frac{1}{N} \sum_j \left\langle \mathcal{R} \mathcal{F}^{-1} \left(\mathcal{F} \{f(t)\} \mathcal{F}^* \{f(t)\} \right) \right\rangle_t, \quad f(t) = e^{im\theta(t)}$$



In the hexatically ordered phase ($k_B T/\epsilon < 0.40$), C_6 no longer decays to zero, and it can be shown that $C_6(\infty)$ instead approaches the square of the global orientational order parameter.

Figure 3: Orientational time-correlation functions at 11 temperatures for $N = 300$, and $\rho^* = 0.25$. Time correlations up to $10^4\tau$ with configurations saved every 10 timesteps were computed. The plot is truncated at $t/\tau = 500$ for clarity. The computed relaxation times, τ_{relax} , are also used to verify simulated relaxation time such that sampled configurations are not influenced by cooling history.

An Isotropic→Hexatic Molecular-Orientational Phase Transition

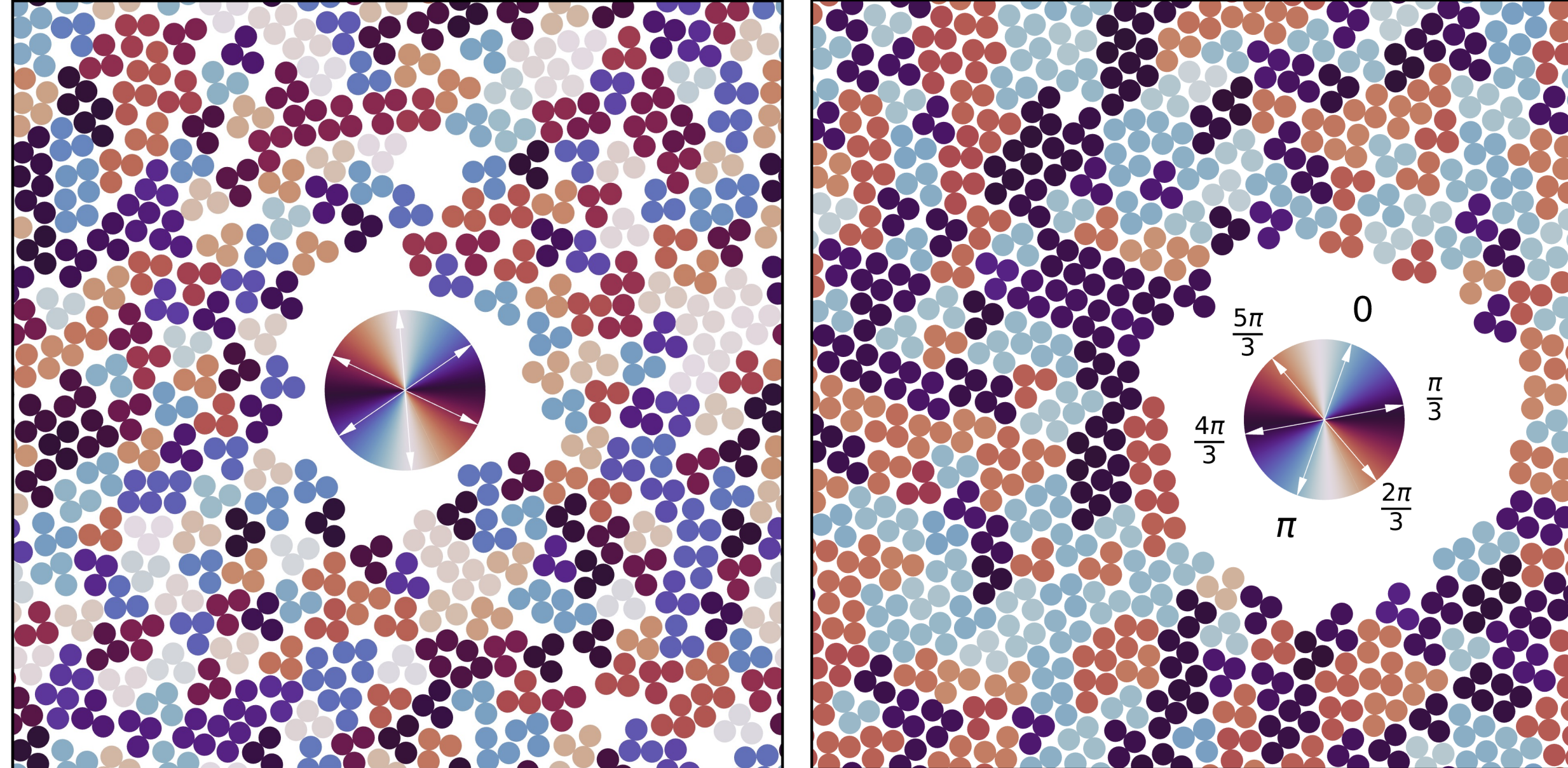


Figure 4: Images of a 2dLW liquid with $N = 300$ at temperatures $k_B T/\epsilon = 0.460$ (left) and $k_B T/\epsilon = 0.224$ (right). Trimers are shaded by orientation relative to the director (the eigenvector corresponding to the greatest eigenvalue of $\underline{Q}^{(6)}$ in each state). Isotropic character dominates on the left, while conspicuous hexatic ordering is present on the right.

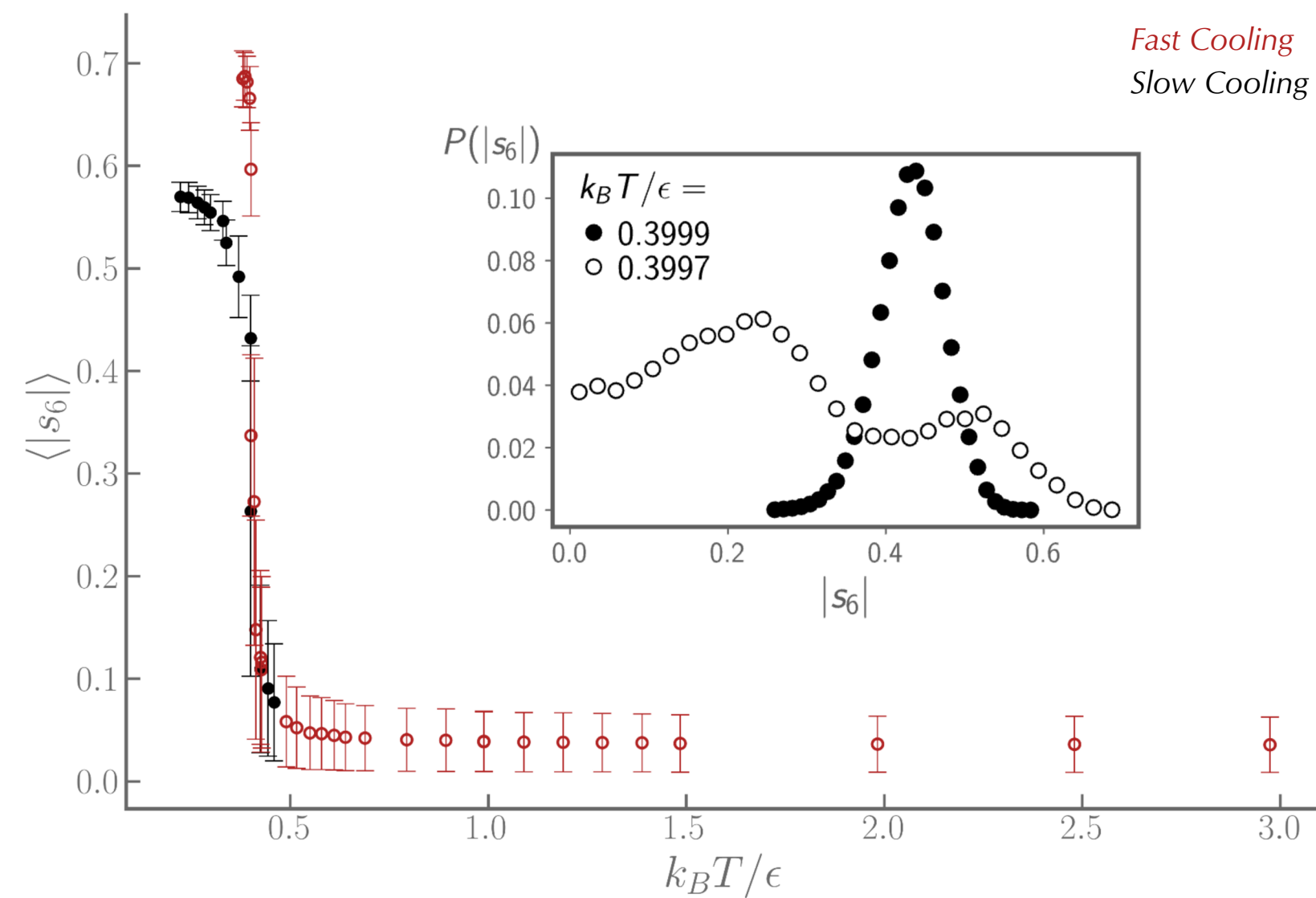


Figure 5: Above: Time-averaged sixfold molecular orientational order parameters ($m = 6$) for a system of $N=300$ 2dLW trimers at a range of temperatures. The error bars traverse ± 1 standard deviation from the mean. Averages are taken over 2×10^6 configurations following relaxation for $2 \times 10^4\tau$. Inset contains probability densities of the sixfold order parameter around the transition.

We characterize the slowing of liquid dynamics using the temperature-dependence of the self-diffusion constant, $D = \frac{1}{4} \frac{\partial \Delta r^2(t)}{\partial t}$. We observe transitions from ‘normal’ liquid dynamics at high temperatures ($k_B T/\epsilon > 1.5$) to a fragile supercooled liquid (fitting the VFT equation), and subsequently to a strong supercooled liquid (fitting the Arrhenius equation at $k_B T/\epsilon \cong 0.388$). Exactly whether the coincident emergence of hexatic order near the fragile-to-strong transition causes the dynamical slowdown remains an important question.

$$D = \begin{cases} D_0 \exp\left(\frac{A}{T - T_0}\right) & \text{Vogel - Fulcher} \\ D_0 \exp\left(\frac{A}{T_0}\right) & \text{Arrhenius} \end{cases}$$

Dynamical Heterogeneity

Dynamical heterogeneity is the localization of ‘active’ regions in a supercooled liquid to small domains amidst a nearly-static background. We observe dynamic heterogeneity in our system, which dominates in the strong-supercooled liquid regime. We measure dynamic heterogeneity using four-point susceptibilities and observe an increase in peak values with decreasing temperature, indicating growing spatial correlations. One objective of ongoing work is to determine to what extent ‘active’ regions of 2dLW below the isotropic-hexatic transition are correlated with orientational ordering, with recent studies on other glass models suggesting that dynamical heterogeneity may have structural origins [5].

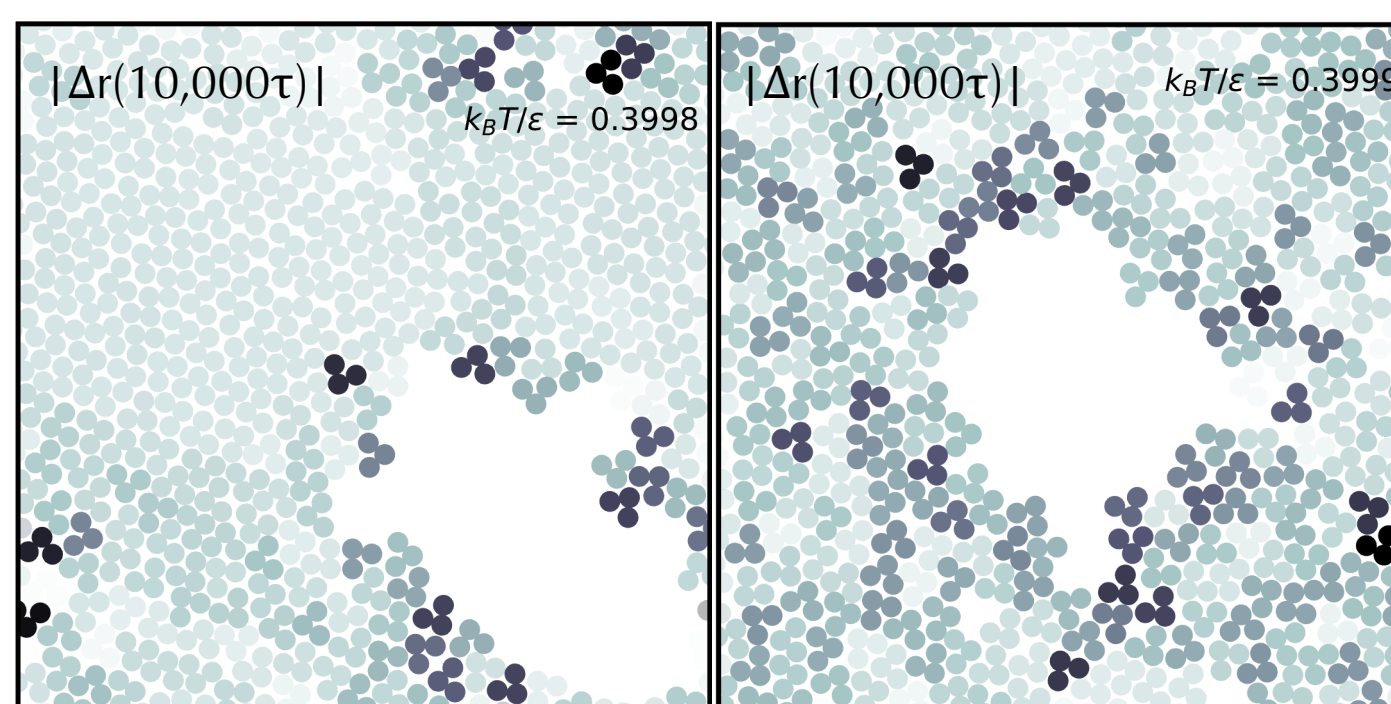


Figure 7: Left: Snapshots of 2dLW liquid below (left) and above (right) the isotropic-hexatic transition, shaded by displacements during a $10,000\tau$ window.

Scaled Particle Theory

It is possible to analytically predict phase transitions in two-dimensional liquids using *Scaled Particle Theory (SPT)*. To do this, we consider trimers composed of hard disks ($V=\infty$ for $r \leq \sigma$) with a 2dLW geometry. The fundamental idea of SPT is that the full equation of state for a 2D liquid can be extracted by computing the free energy cost associated with adding one additional molecule to a packed system at some orientation [3].

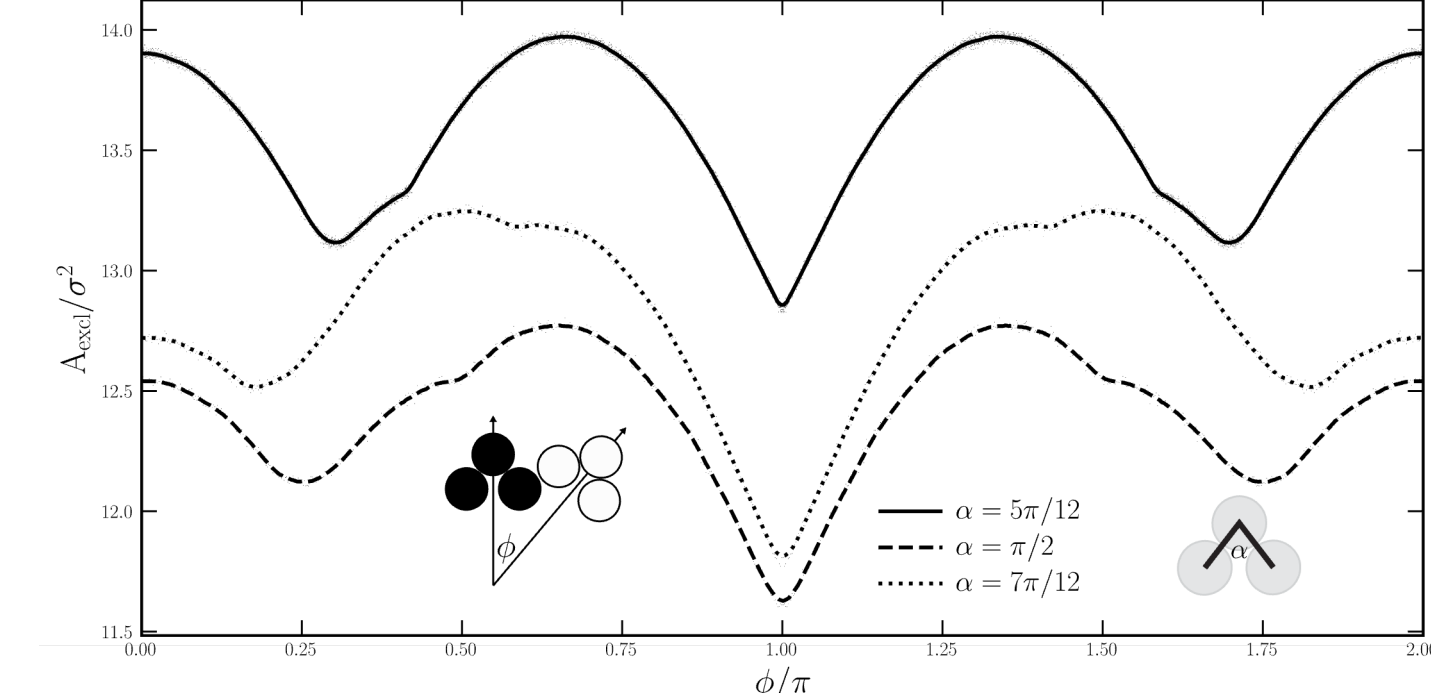


Figure 8: Excluded area as a function of ϕ for 2dLW trimers via Monte Carlo integration, using 10^6 samples with a resolution of 10^{-4} radians for $\alpha = 5\pi/12$ (2dLW), and 2×10^{-2} radians for other apical angles. Solid, dashed and dash-dotted lines are averages over a window of 20 values.

A Simple Preliminary Argument

For density ρ , orientational distribution $f(\Phi)$, relative orientation Φ , free energy per molecule ϕ , packing fraction η , and molecular area a :

$$\rho(\phi) = \rho f(\phi), \quad \eta = \rho a, \quad A^{\text{SPT}}(\phi) = A^{\text{excl}}(\phi) - 2a$$

The preferred $f(\Phi)$ is that with lowest free energy, and minimizes the free energy functional ϕ :

$$\phi[f(\phi)] = \int_0^{2\pi} f \ln(f) d\phi - \ln(1-\eta) + \frac{\rho}{1-\eta} \mathcal{K}[f], \quad \mathcal{K}[f] = \frac{1}{2} \int d\phi d\phi' f(\phi) f(\phi') A^{\text{SPT}}(\phi - \phi')$$

Rewriting in terms of Fourier coefficients, omitting constant terms:

$$A^{\text{SPT}}(\phi) = \sum_k \cos(k\phi), \quad s_k = \frac{1}{2\pi} \int_0^{2\pi} A^{\text{SPT}}(\phi) \cos(k\phi) d\phi, \quad f(\phi) = \frac{1}{2\pi} \left[1 + \sum_{k \geq 1} f_k \cos(k\phi) \right], \quad f_0 = 1, \quad \mathcal{K}[f] = \frac{1}{2} \sum_{k \geq 0} s_k f_k^2$$

It can be shown from the second variation of ϕ that that the isotropic phase is no longer stable at sufficient η . Then, the free energy is minimized when s_k is *most negative*, with the corresponding coefficient f_k maximized. Computing s_k for the Monte-Carlo derived 2dLW A^{excl} , s_k is most negative for $k = 6$. So, $f(\Phi)$ ought to be dominated by f_6 – which possesses sixfold rotational symmetry.

In the limit of $N \rightarrow \infty$ we argue that this is sufficient to induce hexatic orientational order.

Ongoing Efforts: Monte-Carlo Sampling

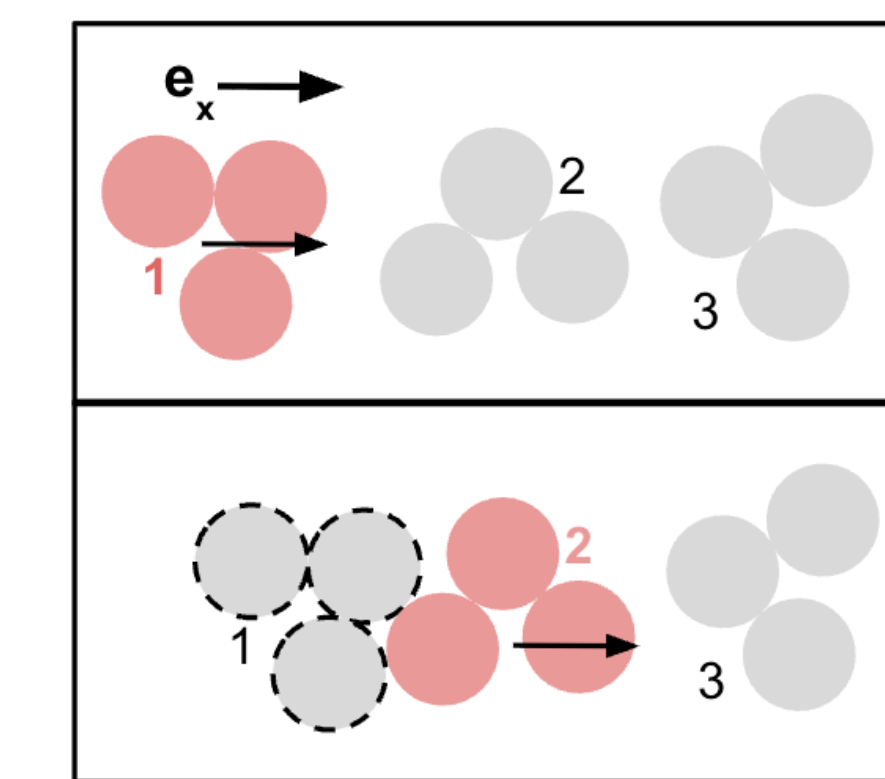


Figure 10: Translational motion is simulated by choosing a random molecule (1) and displacing it along a chosen axis (traditionally only in the positive direction). When a collision occurs, a ‘lifting’ move is performed such that molecule 2 now continues in the same direction. This continues until the total displacement reaches a chosen chain length l , which we choose to be $l = 0.2L$ where L is the size of the MD simulation box from which the initial configuration is taken.

Conclusions & Outlook

Thus far, we have compelling evidence that the slow dynamics of a supercooled 2dLW liquid are coupled with the emergence of hexatic molecular orientational order, and preliminary calculations using SPT suggest that hexatic molecular orientational order in 2dLW is a direct result of the model’s geometry. We show that hexatic ordering dominates below the transition from ‘normal’ to supercooled liquid, but further study is required to determine its relation to dynamics at temperatures approaching the glassy regime.

Ongoing work involves sampling relaxed configurations of our system in the deeply-supercooled regime by adapting Monte-Carlo techniques shown to dramatically accelerate equilibration of hard-sphere glass models, so that we can better understand the role of hexatic order near and below the glass transition. Furthermore, we anticipate completing MD simulations in the isothermal-isobaric ensemble to eliminate the possibility of spurious effects due to cavitation.

References

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